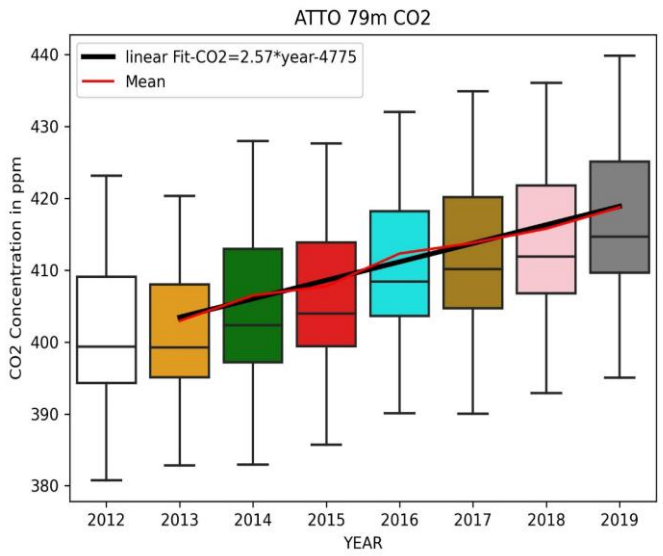
How Weather Events Change the Concentration of Gases in the Atmosphere

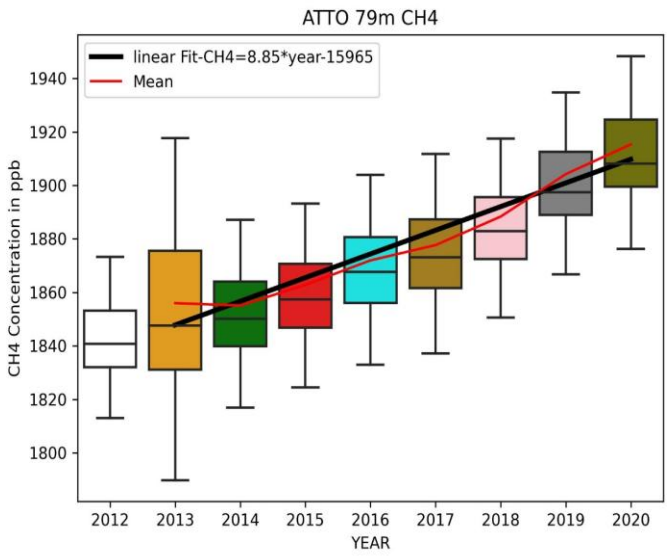
Luiz A. T. Machado, Jonathan Williams, Hartwig Harder and Christopher Pöhlker

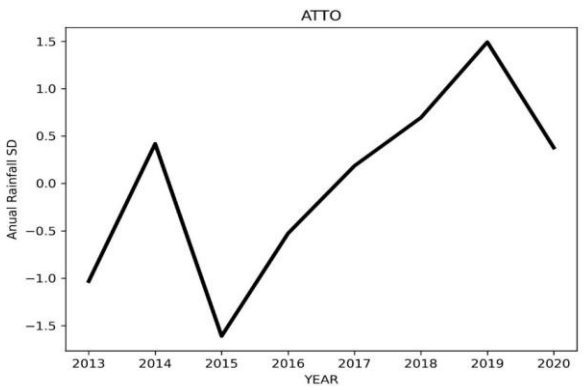
ATTO Green House, Reactive and VOC gases

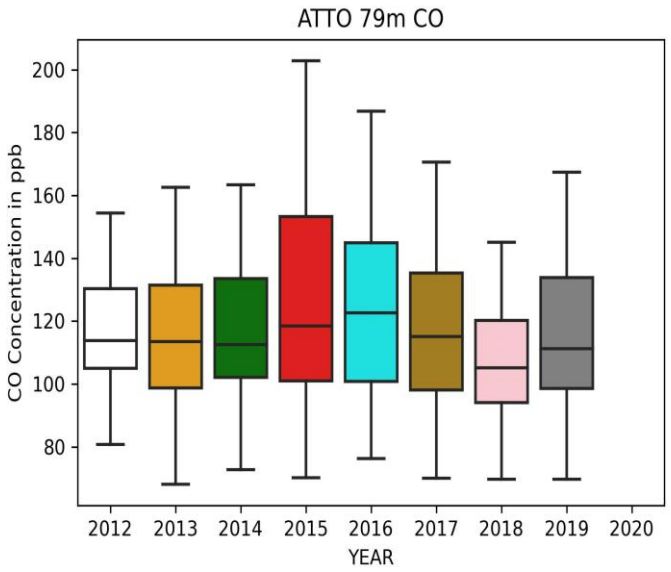
- Interannual
- Seasonal
- Diurnal
- During Rainfall Events

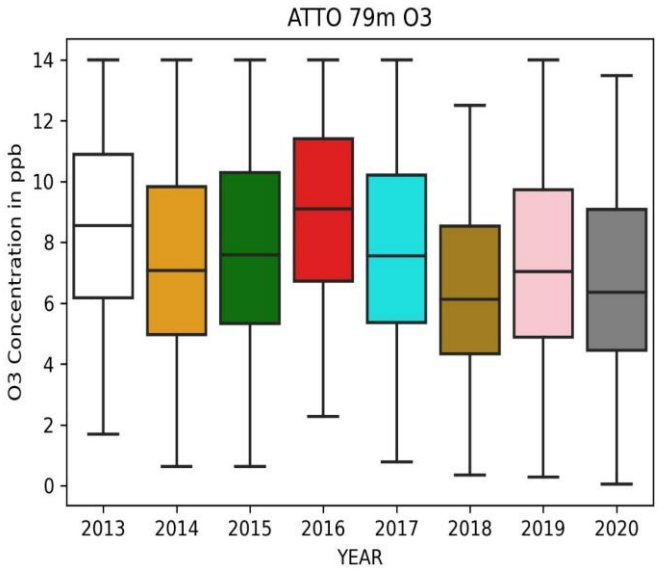
Interannual



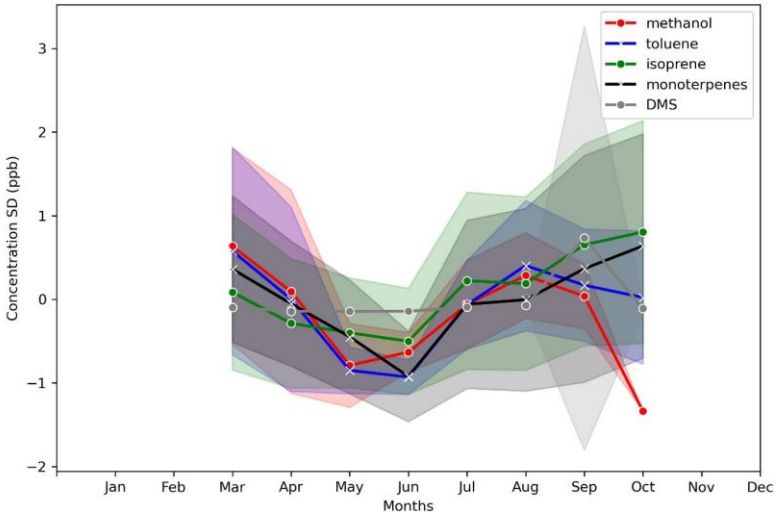


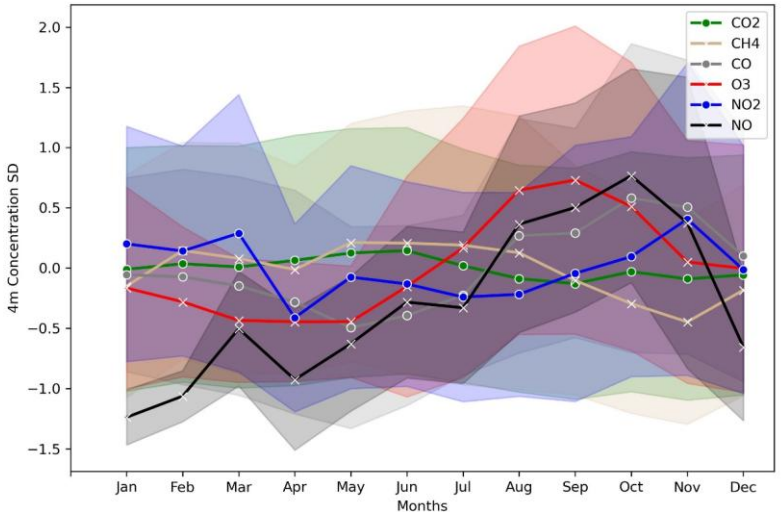


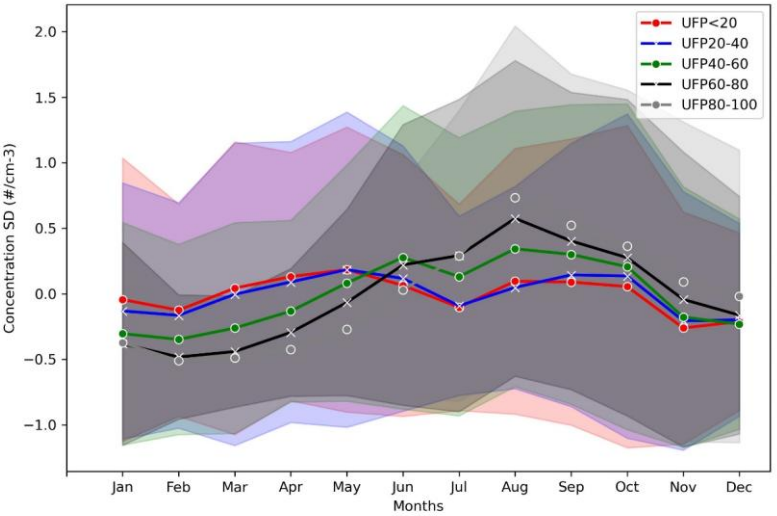


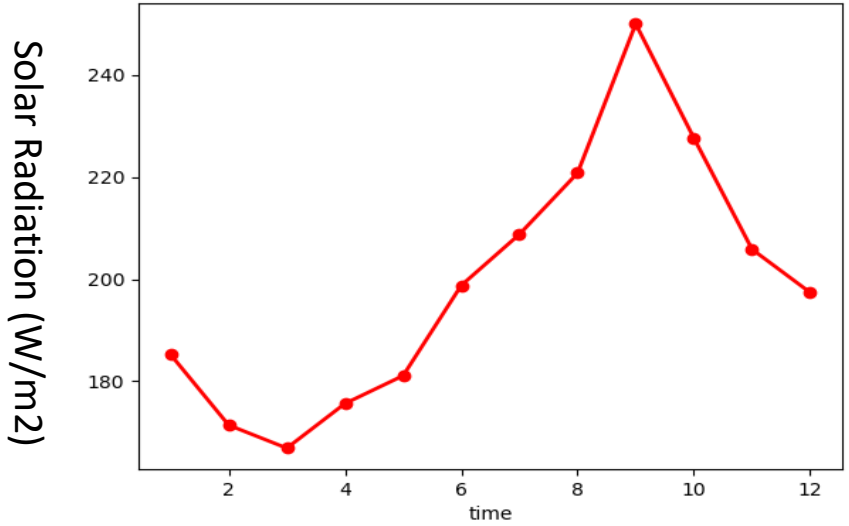


Seasonal

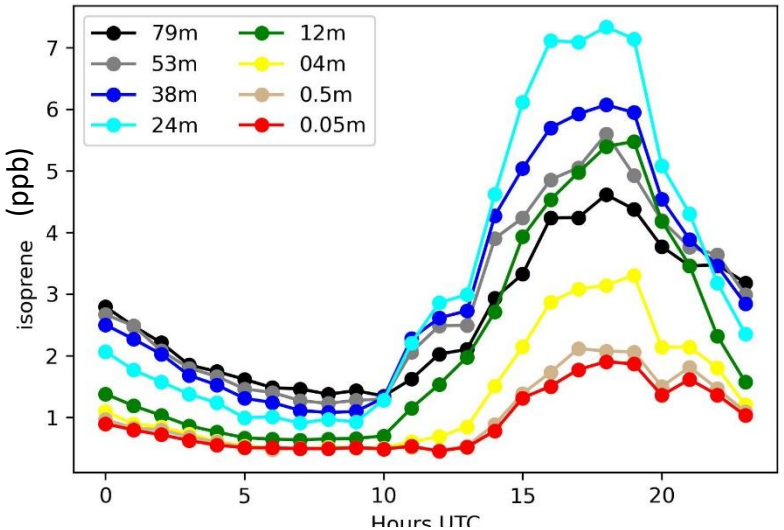

methanol
toluene
isoprene
monoterpenes
DMS
Concentration SD (ppb)
Months
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

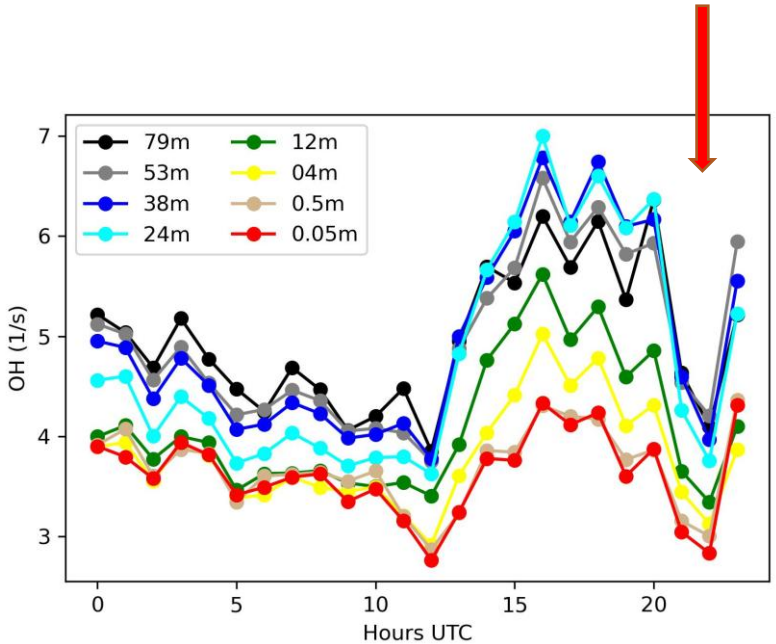

CO2
CH4
CO
O3
NO2
NO
4m Concentration SD
Months
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

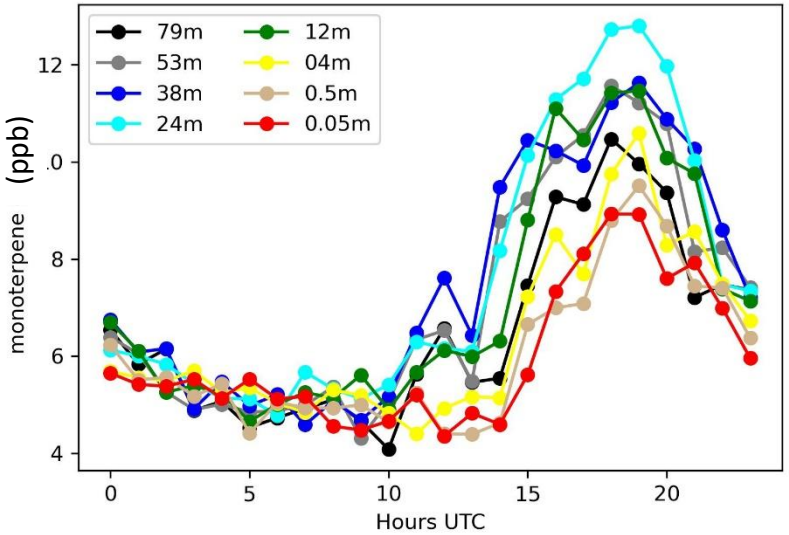

UFP<20
UFP20-40
UFP40-60
UFP60-80
UFP80-100
Concentration SD (#/cm-3)
Months
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

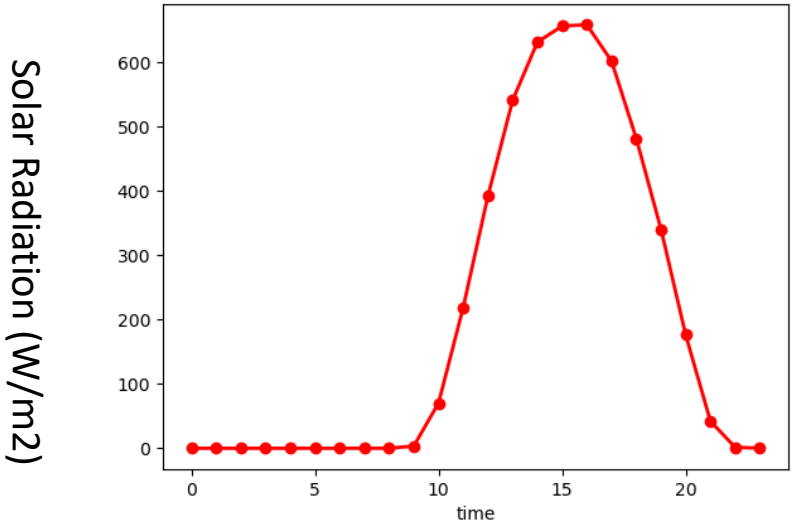

Solar Radiation (W/m2)
time

Diurnal

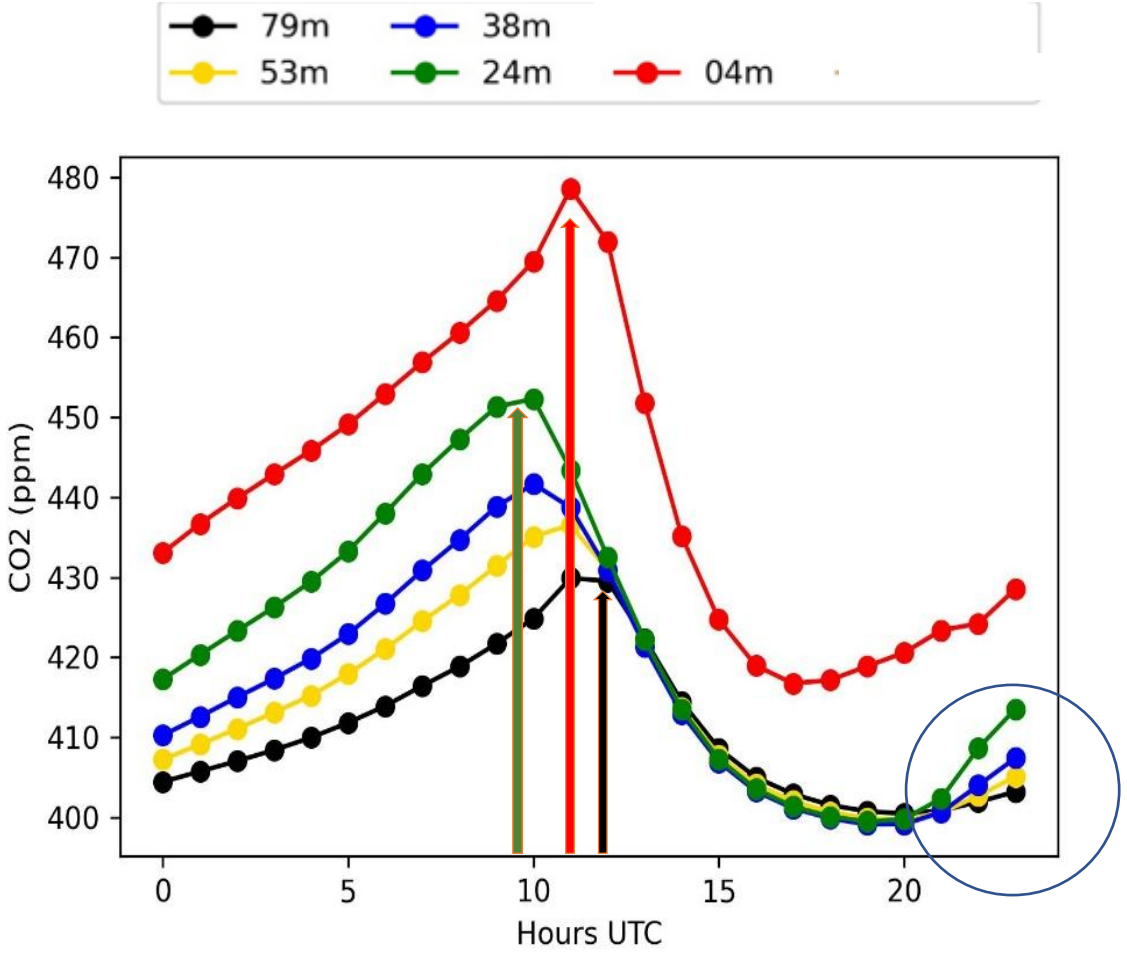

79m
53m
38m
24m
12m
04m
0.5m
0.05m
isoprene (ppb)
Hours UTC

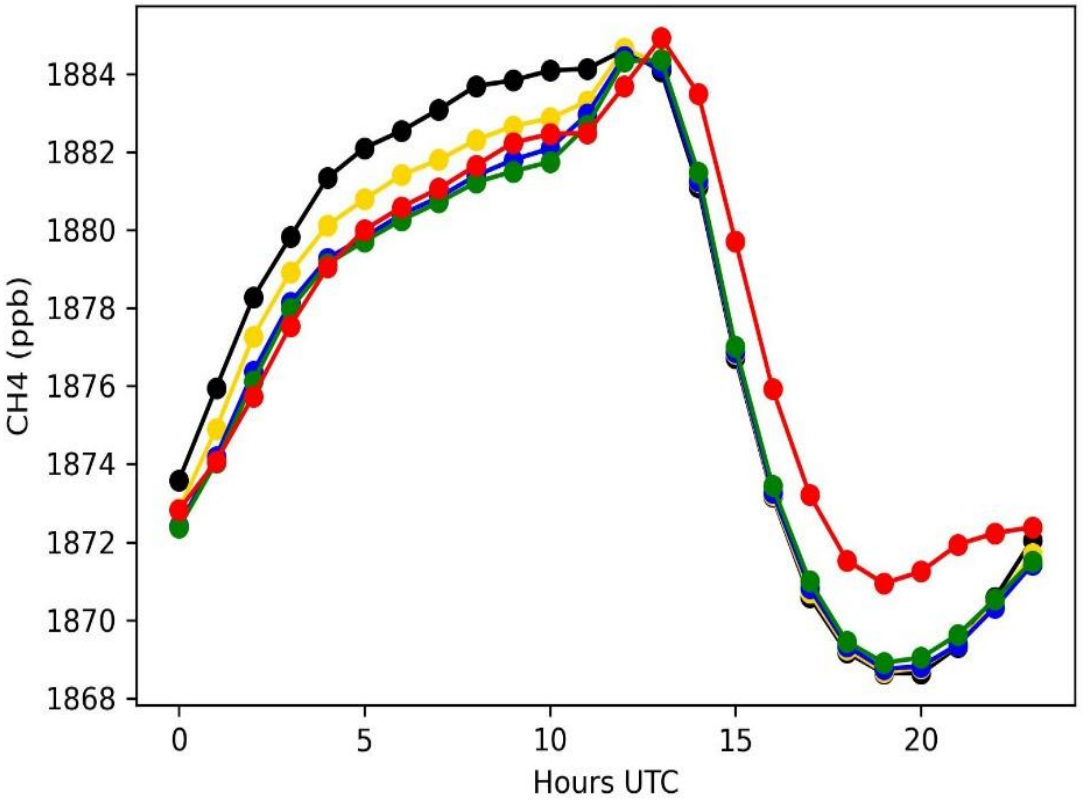

79m
53m
38m
24m
12m
04m
0.5m
0.05m
OH (1/s)
Hours UTC


79m
53m
38m
24m
12m
04m
0.5m
0.05m
monoterpene (ppb)
Hours UTC

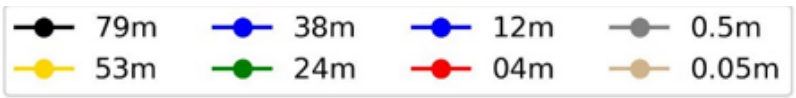

Solar Radiation (W/m2)
time

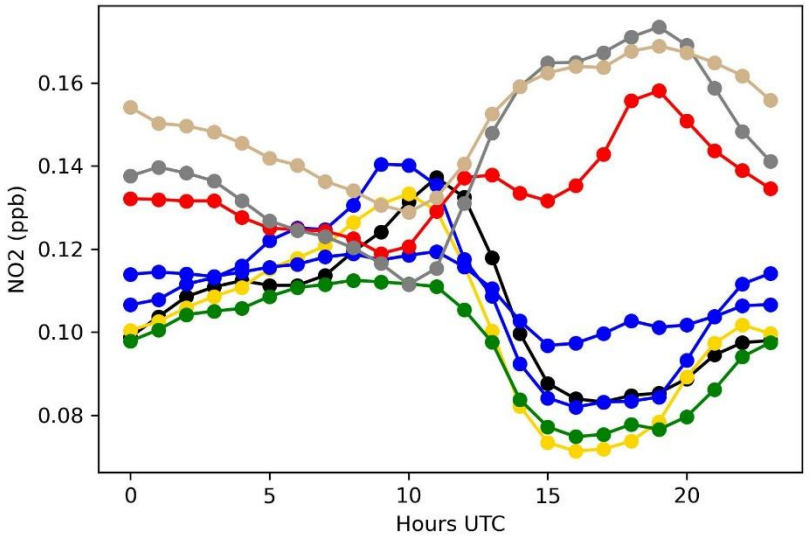
Greenhouse Gases

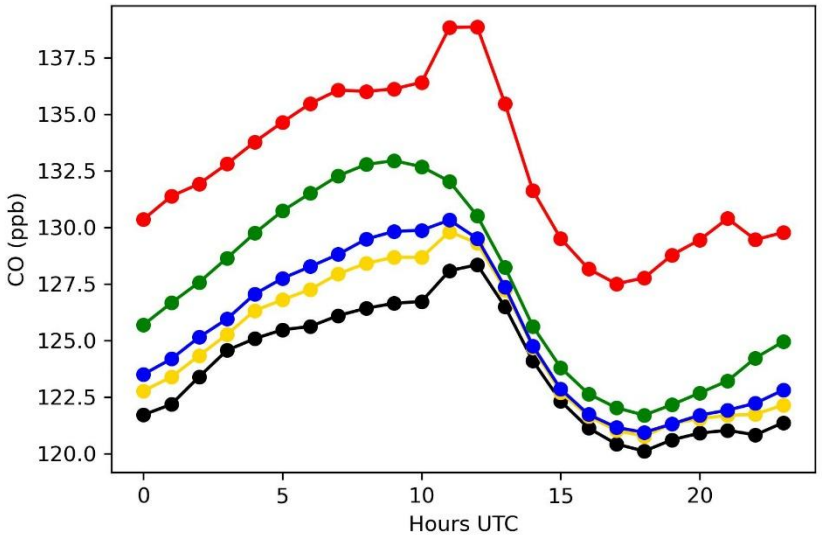


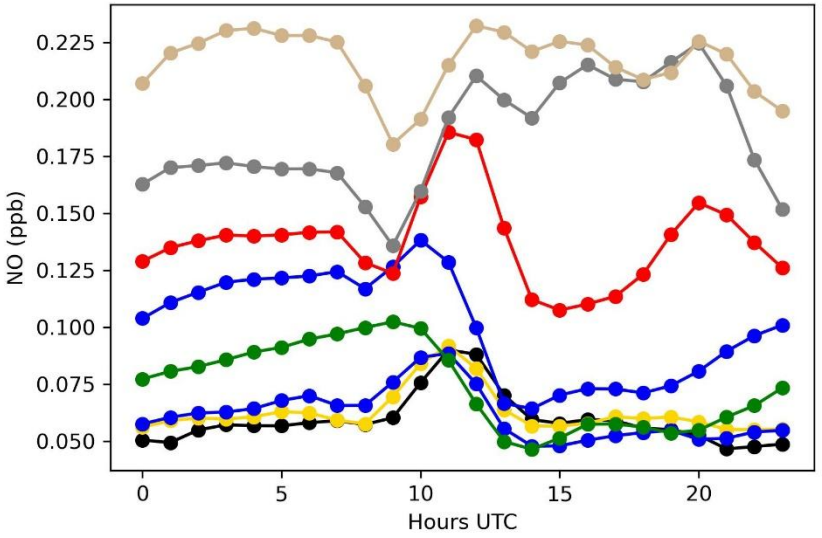


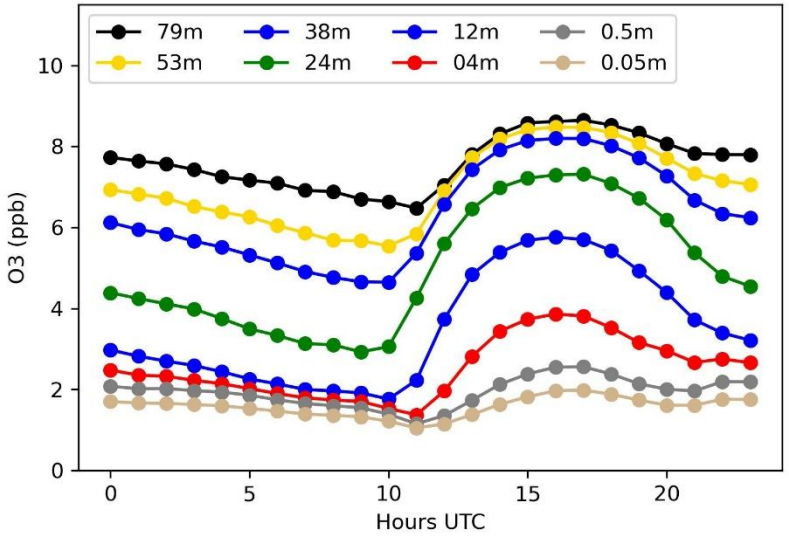
Reactive Gases



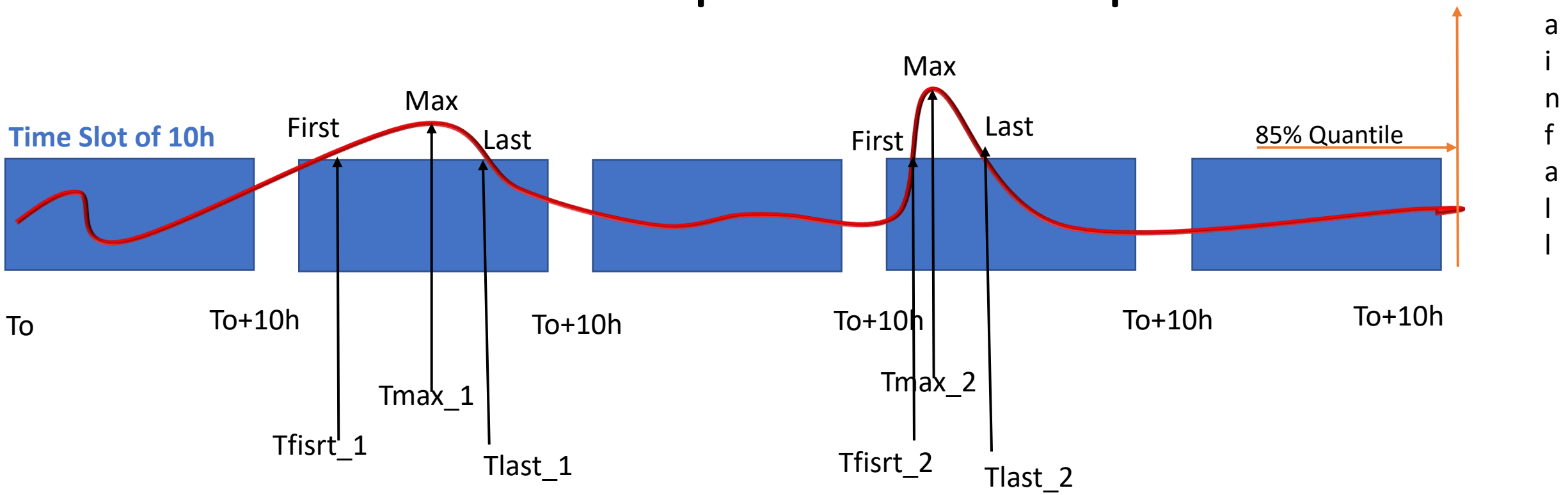








How the Composite was computed



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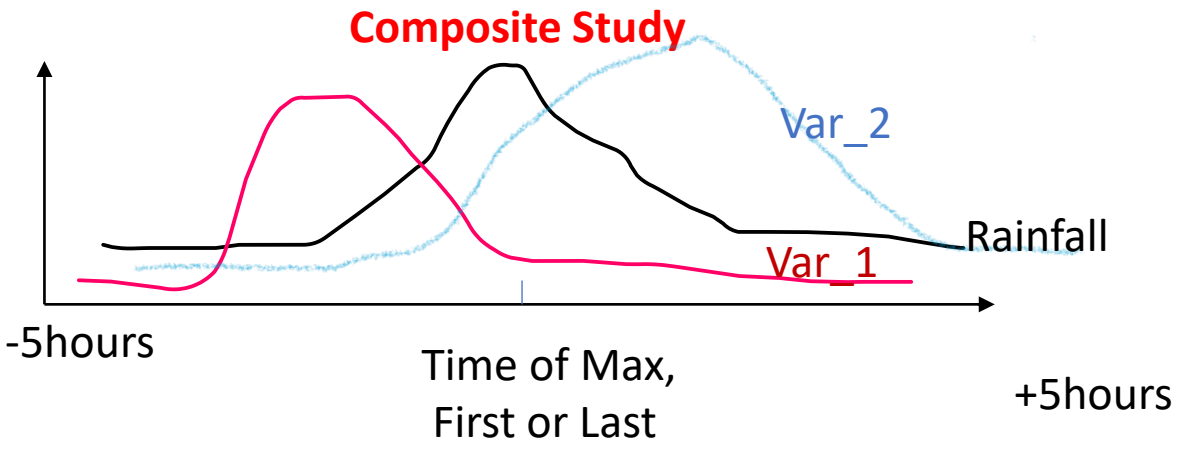
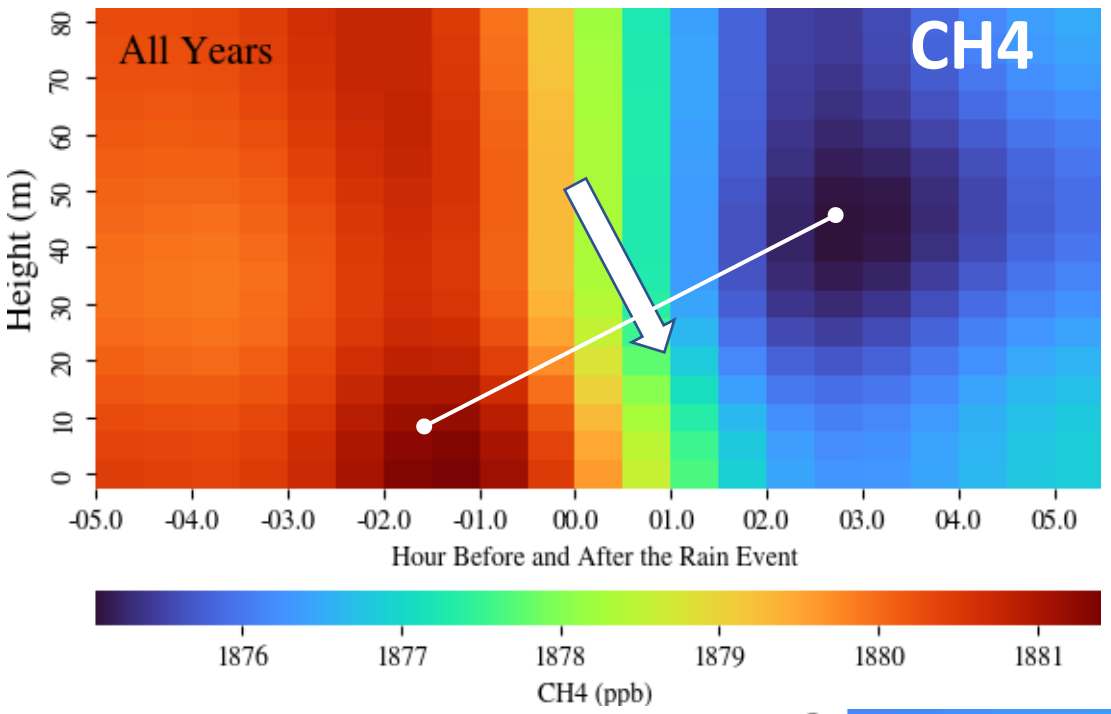
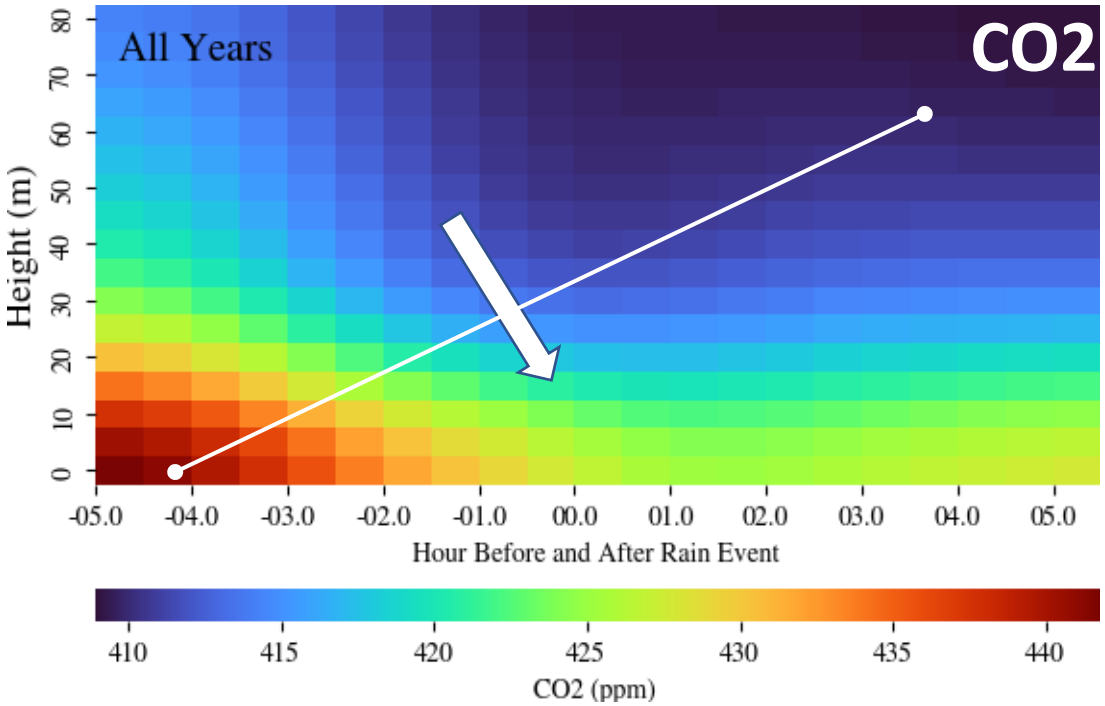
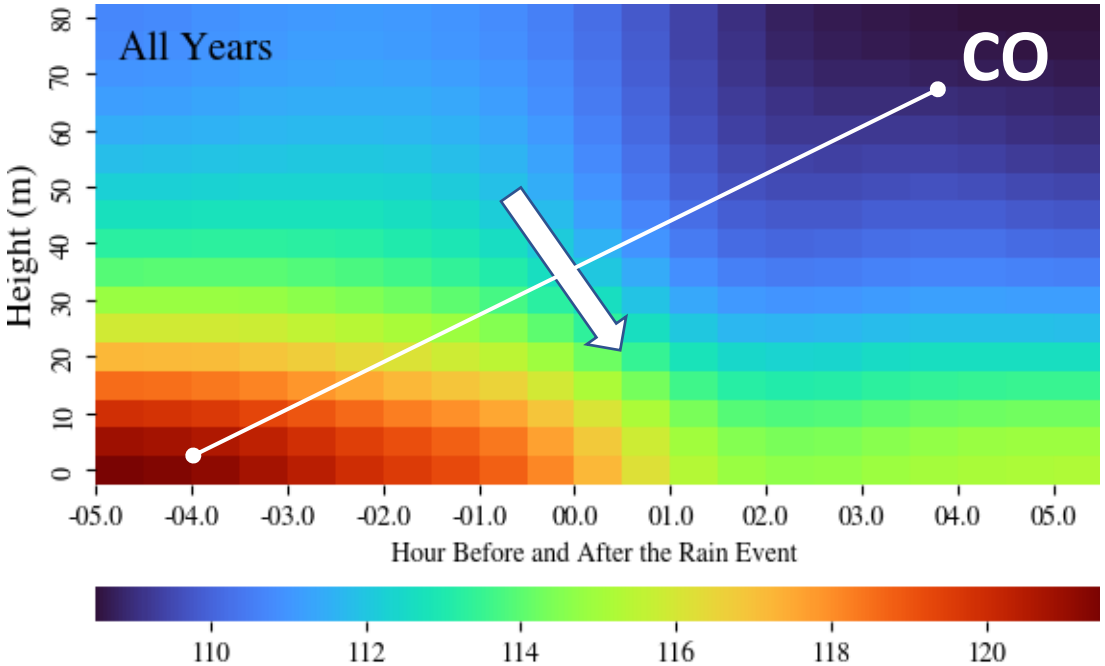
[Tmax_1,Tmax_2, Tmax_3,…Tmax_n]

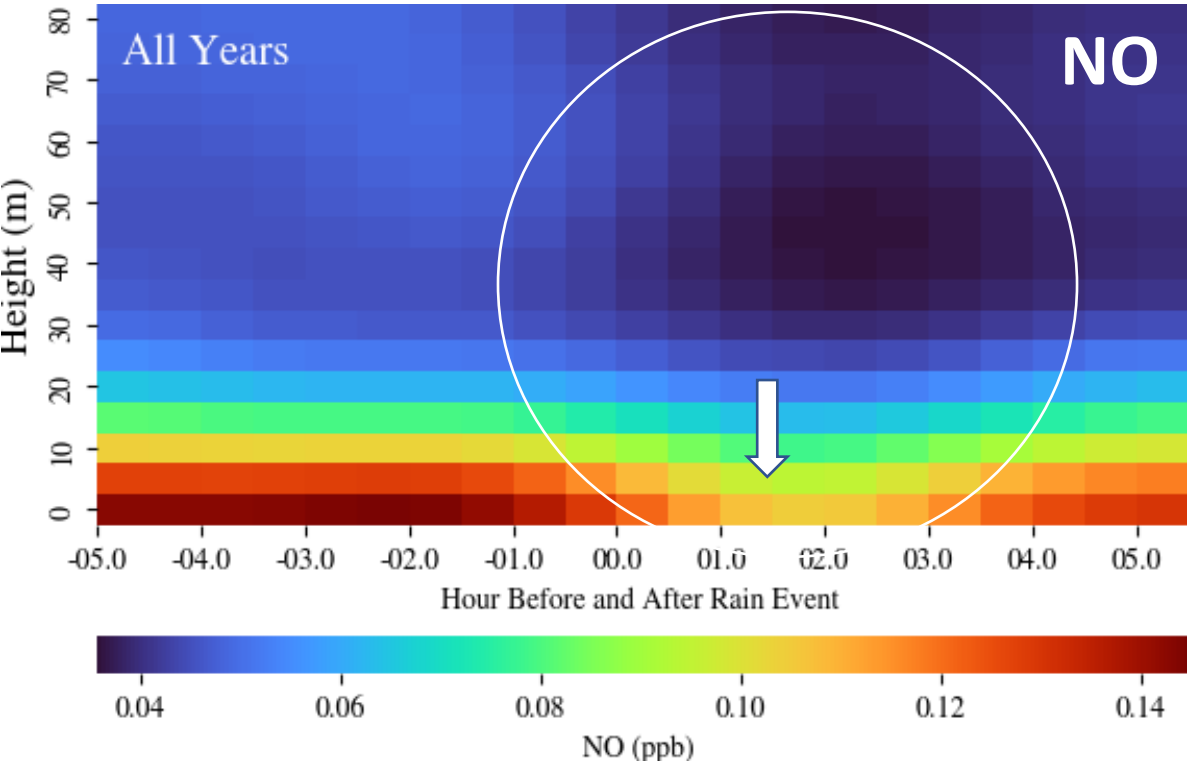
[Tlast_1,Tlast_2, Tlast_3,…Tlast_n]

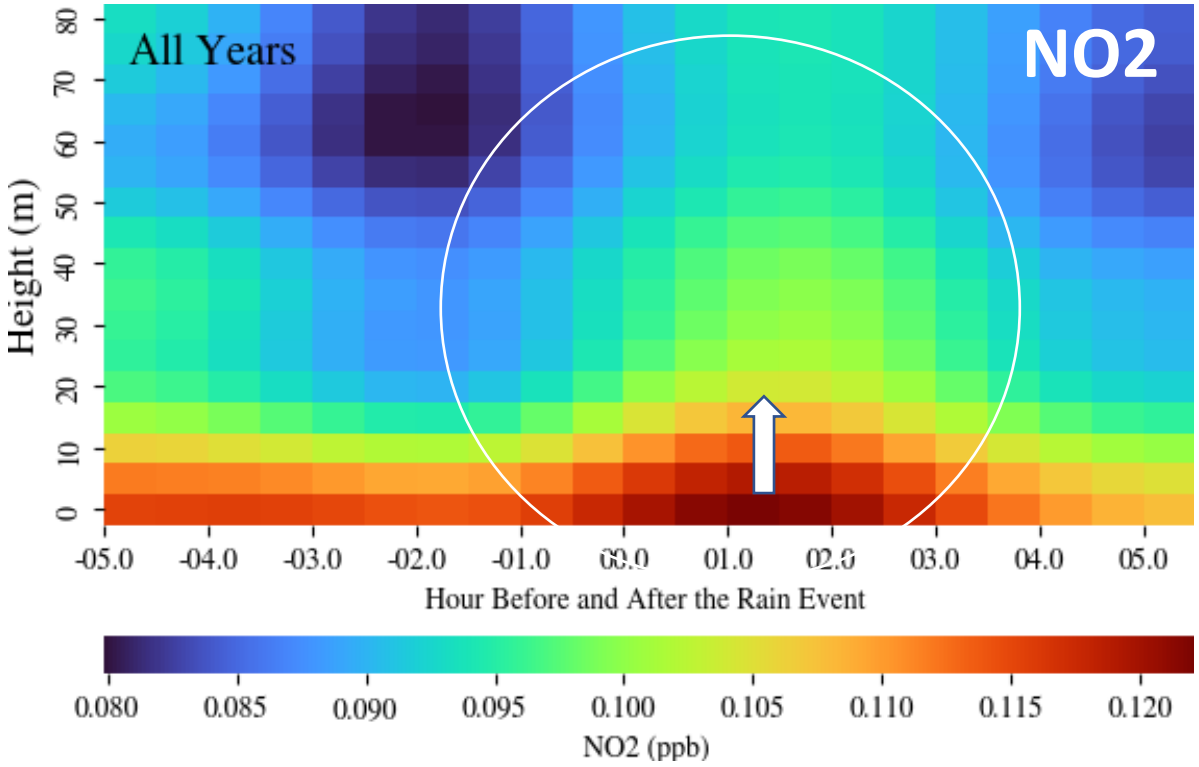


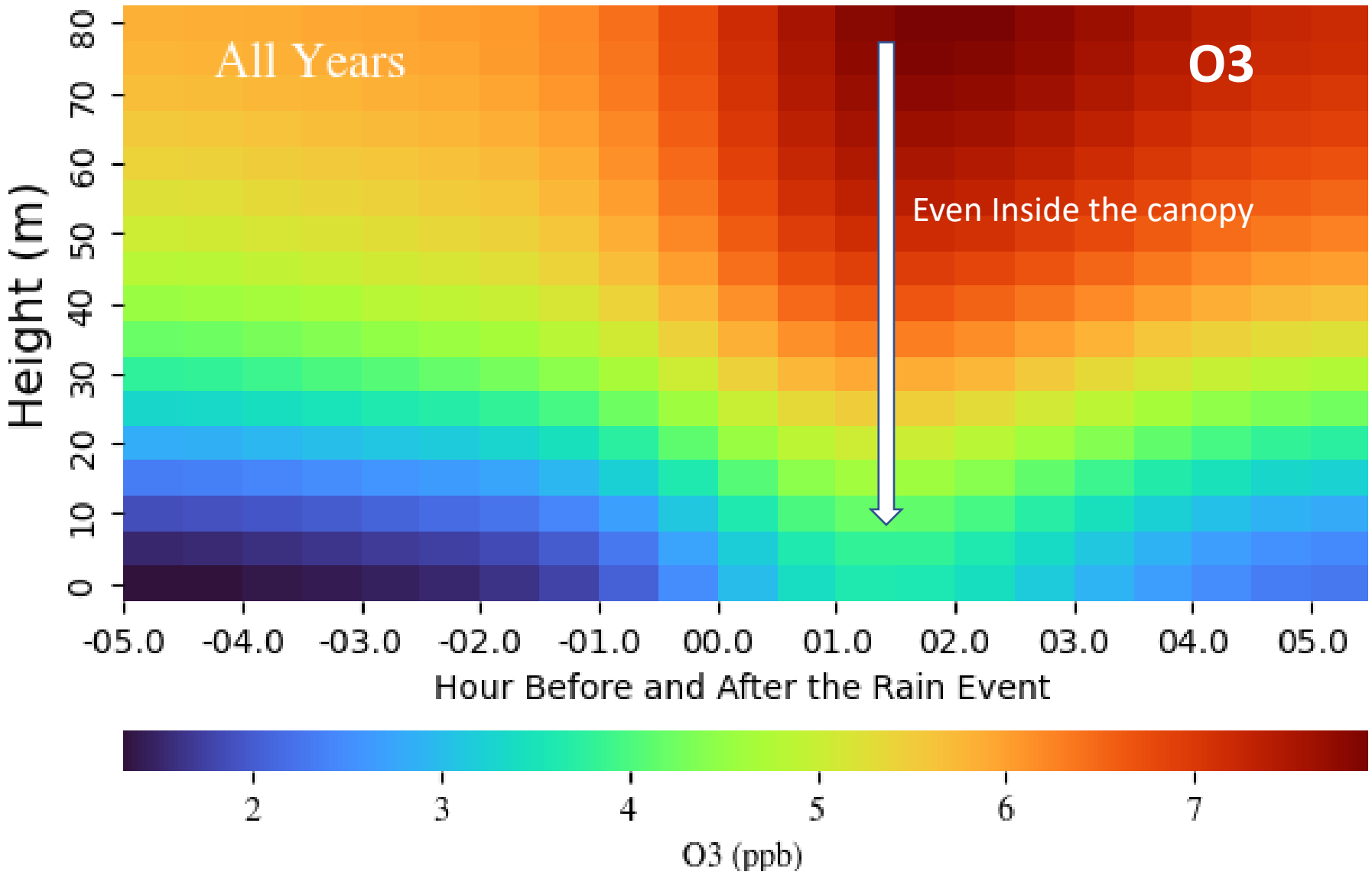

All Years
CH4
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After the Rain Event
1876
1877
1878
1879
1880
1881
CH4 (ppb)


All Years
CO2
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After Rain Event
410
415
420
425
430
435
440
CO2 (ppm)

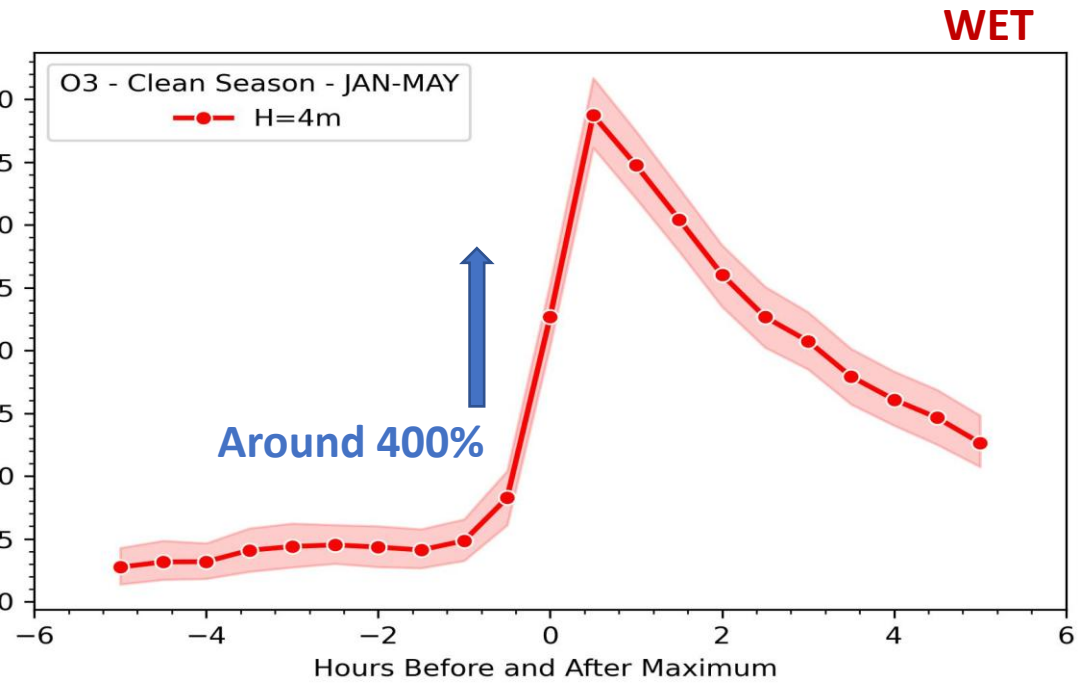

All Years
CO
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After the Rain Event
110
112
114
116
118
120

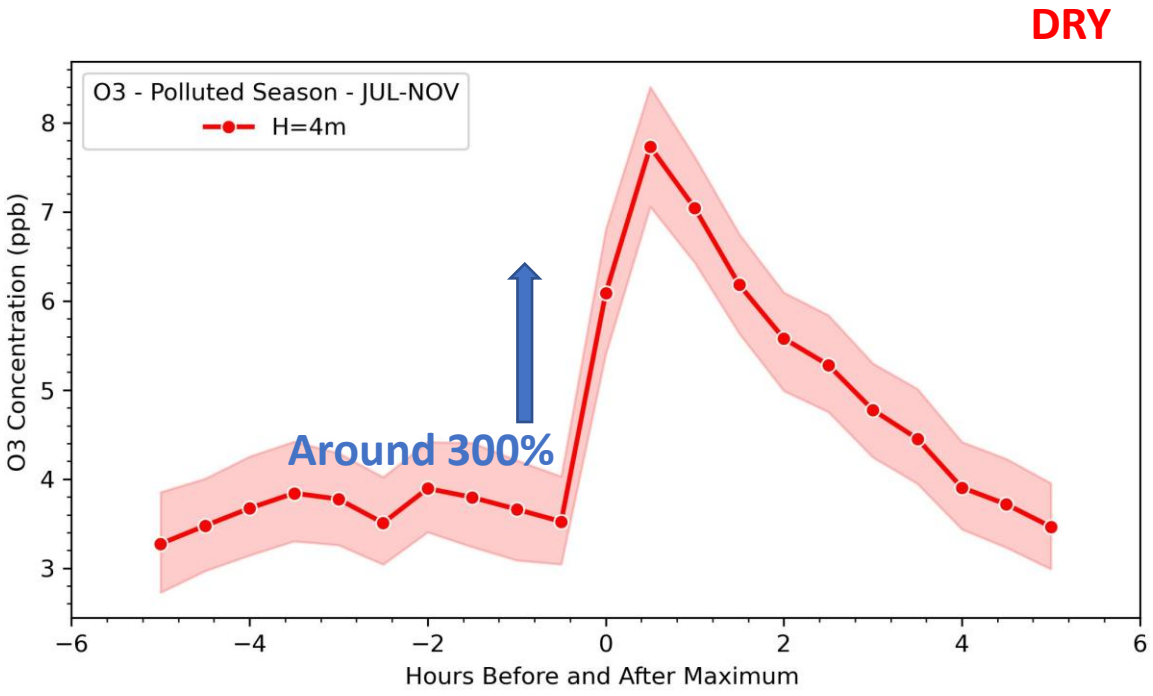

All Years
NO
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After Rain Event
0.04
0.06
0.08
0.10
0.12
0.14
NO (ppb)


All Years
NO2
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After the Rain Event
0.080
0.085
0.090
0.095
0.100
0.105
0.110
0.115
0.120
NO2 (ppb)

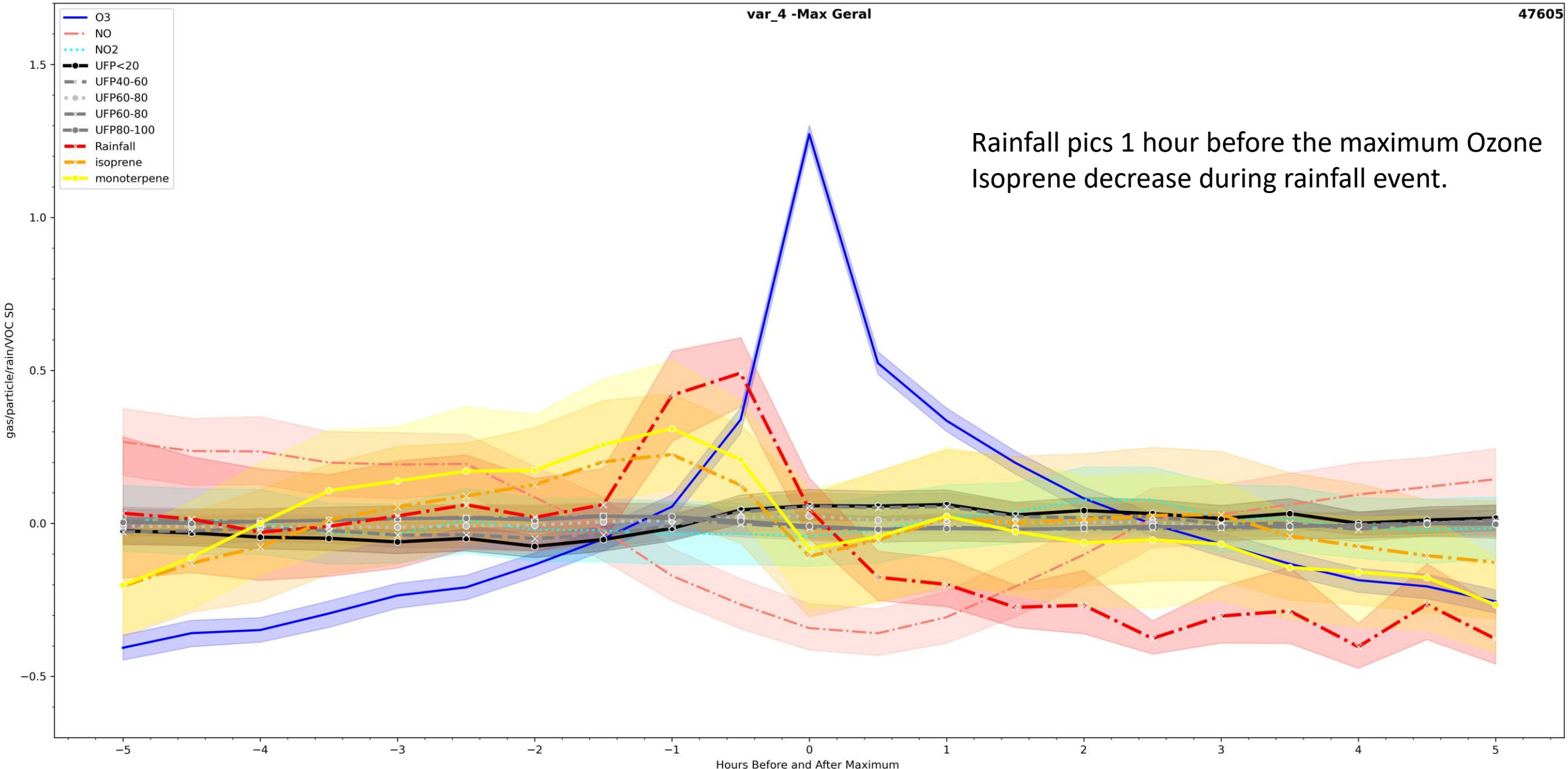

All Years
O3
Even Inside the canopy
Height (m)
0
10
20
30
40
50
60
70
80
-05.0
-04.0
-03.0
-02.0
-01.0
00.0
01.0
02.0
03.0
04.0
05.0
Hour Before and After the Rain Event
2
3
4
5
6
7
O3 (ppb)

Rainfall and O3 Concentration Variation



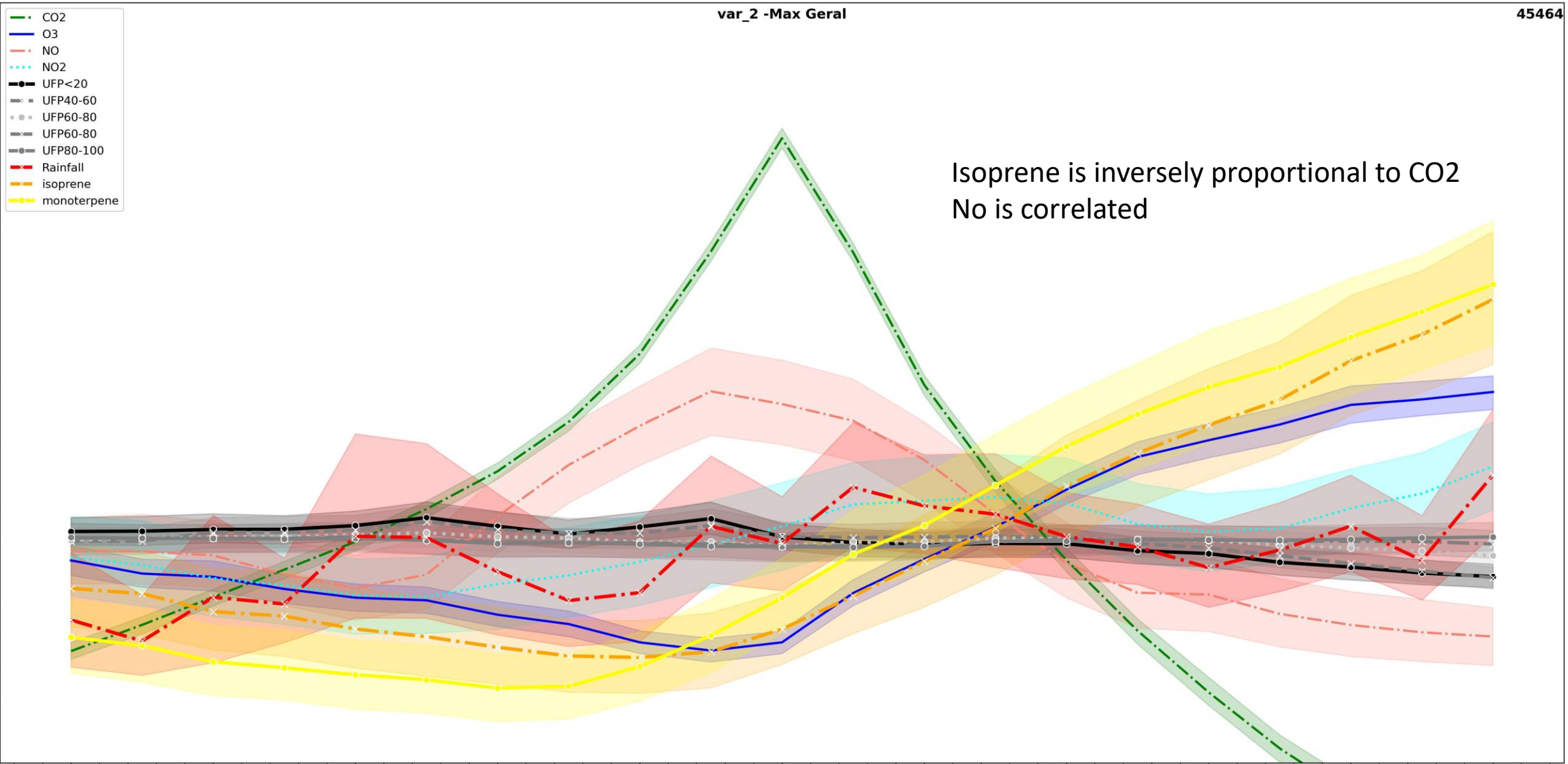


The Composite Study – Ozone

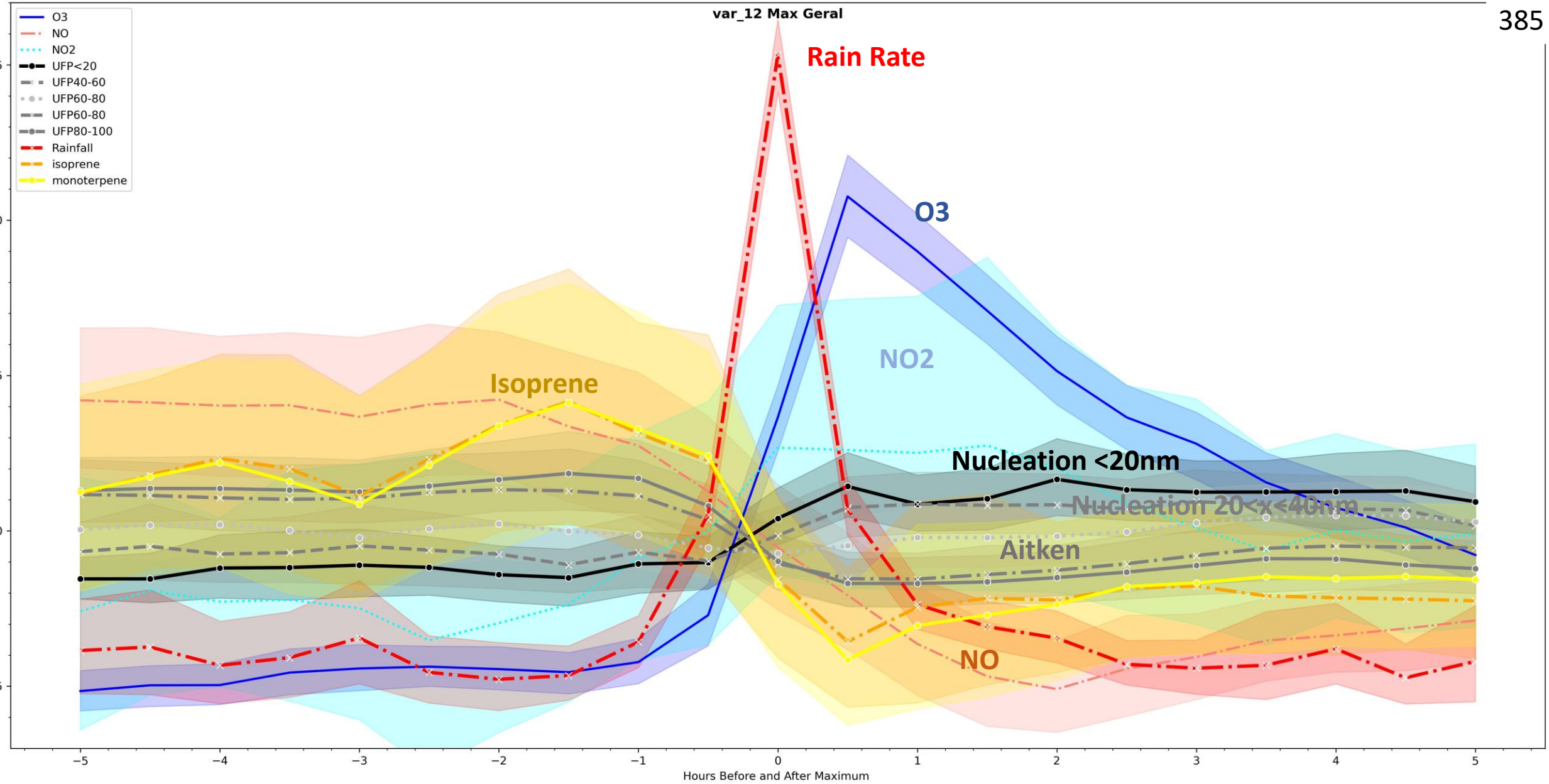


Rainfall pics 1 hour before the maximum Ozone
Isoprene decrease during rainfall event.

The Composite Study – CO2



The Composite Study - Rainfall



Conclusions

There is an intrinsic relationship between greenhouse gases, reactive and VOCs with weather events. Boundary layer evolution, rainfall events change the gas concentration considerably and at the same time these gases react to form new particles that will act as condensation nuclei. The knowledge of this feedback mechanism between rainfall and gas concentration is very important in a planet with a changing climate. The formation of nucleation particles that later will form the cloud condensation nuclei seems to be related to gas reactions, and at the same time rain events modifing the gas concentration.